Practical No. 4

Aim: To implement a Business UDDI Registry using Java.

Theory

UDDI (Universal Description, Discovery and Integration) is a platform-independent registry used to publish and discover web services.

A UDDI Business Registry stores details such as:

- Business Key (Unique ID)
- Business Name
- Description
- Contact Information

In this practical, we simulate a UDDI registry using Java classes.

Step 1: Create a Java Project

Project Name:

UDDIRegistryDemo

Package Name:

uddidemo

Step 2: Create Business.java

```
package uddidemo;

public class Business {

    private String businessKey;

    private String businessName;

    private String description;

    private String contact;

    public Business(String businessKey, String businessName,

                String description, String contact) {
```

```
        this.businessKey = businessKey;
        this.businessName = businessName;
        this.description = description;
        this.contact = contact;
    }
    public String getBusinessKey() {
        return businessKey;
    }
    public void displayBusiness() {
        System.out.println("--------------------------------");
        System.out.println("Business Key : " + businessKey);
        System.out.println("Business Name: " + businessName);
        System.out.println("Description  : " + description);
        System.out.println("Contact      : " + contact);
        System.out.println("--------------------------------");
    }
}
```

Step 3: Create Registry.java

```
package uddidemo;
import java.util.ArrayList;
public class Registry {
    ArrayList<Business> businessList = new ArrayList<>();
    // Register Business
    public void registerBusiness(Business business) {
        businessList.add(business);
        System.out.println("Business Registered Successfully.");
    }
    // Display All Businesses
    public void displayBusinesses() {
        System.out.println("\n===== UDDI BUSINESS REGISTRY =====");
        for (Business business : businessList) {
            business.displayBusiness();
        }
    }
    // Search Business
    public void searchBusiness(String key) {
        boolean found = false;
        for (Business business : businessList) {
            if (business.getBusinessKey().equalsIgnoreCase(key)) {
                System.out.println("\nBusiness Found");
```

```
                business.displayBusiness();
                found = true;
                break;
            }
        }
        if (!found) {
            System.out.println("\nBusiness Not Found.");
        }
    }
}
```

Step 4: Create Main Class (UDDIRegistryDemo.java)

```
package uddidemo;
public class UDDIRegistryDemo {
    public static void main(String[] args) {
        Registry registry = new Registry();
        Business b1 = new Business(
                "BUS101",
                "ABC Technologies",
                "Software Development Company",
                "abc@gmail.com");

        Business b2 = new Business(
                "BUS102",
                "XYZ Solutions",
                "Cloud Service Provider",
                "xyz@gmail.com");

        Business b3 = new Business(
                "BUS103",
                "PQR Infotech",
                "Web Application Development",
                "pqr@gmail.com");
```

```
        // Register Businesses
        registry.registerBusiness(b1);
        registry.registerBusiness(b2);
        registry.registerBusiness(b3);
        // Display Registry
        registry.displayBusinesses();
        // Search
        registry.searchBusiness("BUS102");
    }
}
```

Output:

